

PTCy Meta-Analysis: Post-Block-9 Reanalysis Summary

251 Studies · 18 Bayesian Models · 3 Sensitivity Analyses

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i Note

This report was generated using AI under general human direction. At the time of generation, the contents have not been comprehensively reviewed by a human analyst.

1 Executive summary

This report summarises the post-Block-9 reanalysis of the PTCy meta-analysis, expanding the corpus from 220 to 251 studies (525 arms, 14 RCTs). Eighteen Bayesian random-effects models were fitted across three comparisons and six outcome domains, plus three C2 sensitivity models with expanded arm inclusion.

Key findings:

- **C1 OS** and **C1 CMV** are stable: OR 0.79 [0.73–0.85] and 1.26 [1.07–1.47] respectively, consistent with pre-Block-9.
- **C1 aGVHD** (OR 0.67 [0.59–0.78]) shows significant funnel asymmetry (Egger $p = 0.041$); trim-and-fill adjusted OR crosses null. GRADE downgraded to VERY LOW.
- **C2 CMV** shifted from OR 0.77 [0.60–0.99] to 0.92 [0.73–1.15] — CrI now crosses null. However, the C2 sensitivity (PTCy+ATG combo arms included, $k = 22$) shows CMV converging to OR 1.00, confirming the T-cell depletion hierarchy.
- **Arm classification errors** were identified in three Block-9 studies (Sayin, Xu, Zu); these were corrected in the workbook.
- **C3 models** (within-PTCy dose/schedule) are all null at current sample sizes.
- **CMV serostatus** analysis found no significant moderating effect of recipient seropositivity or D+/R+ prevalence; the D–/R+ stratum was the only one showing a neutral PTCy effect ($k = 1$, hypothesis-generating).

2 Database expansion

Table 1: Database expansion from pre-Block-9 (220 studies) to post-Block-9 (251 studies).

Sheet	Pre-Block-9	Post-Block-9	Delta
Studies	220	251	+31
Arms	457	525	+68
Outcomes	3050	3604	+554
Outcomes (subgroup)	116	126	+10
ROB assessments	210	241	+31
RCTs	10	14	+4
C1 eligible arms	301	313	+12
C2 eligible arms	101	103	+2*
C3 eligible arms	29	68	+39

* C2 arms net of 6 reclassified arms (Sayin, Xu, Zu).

New RCTs from Block 9 include Koura 2025 (PTCy + abatacept), Zhang 2023 (reduced-dose PTCy + low-dose ATG), Zu 2022 and Zu 2023 (low-dose PTCy + ATG RCTs).

3 Methodological notes

3.1 Build-analytic corrections

The original `build_analytic()` function back-calculated event counts from cumulative incidence percentages when `event_count` was missing. This inflated `k` by 3–4× for some outcomes (e.g., C1 aGVHD: `k` = 81 instead of `k` = 28). The corrected function **requires actual event_count**, matching the pre-Block-9 methodology.

3.2 Model specification

All models include a `tp_early` covariate (0 = preferred timepoint, 1 = fallback) matching the pre-Block-9 specification. The general model form is:

$$\begin{aligned} \text{events}_i \mid \text{trials}(n_i) &\sim \text{Binomial}(\pi_i) \\ \text{logit}(\pi_i) &= \beta_0 + \beta_1 \cdot \text{ptcy}_i + \beta_2 \cdot \text{tp_early}_i + u_{s[i]} \\ u_s &\sim \mathcal{N}(0, \tau^2) \end{aligned}$$

M2 models add a steroid-exposure covariate using **complete-case analysis** (arms with missing `steroid_pct` dropped), replacing the pre-Block-9 `brms::mi()` approach that is no longer supported in `brms` v2.23+.

3.3 Arm reclassification

Three Block-9 studies had arm classification errors affecting C2 eligibility:

Table 2: Arm reclassification corrections.

Study	Arm IDs	Problem	Action
Sayin 2024 [348]	387, 388	Comparator is CNI+MTX (MRD), not ATG	C2 → N
Xu ZL 2025 [396]	442, 443	PTCy arm = PTCy+ATG(low) combo	C2 → N
Zu Y 2022 [464]	522, 523	PTCy arm = low-dose PTCy+ATG combo	C2 → N

4 C1 results: PTCy vs CNI+MTX/MMF

Table 3: Comparison 1 primary results (post-Block-9). M2 = steroid-adjusted model.

Outcome	Model	k	N	OR [95% CrI]	
OS	M1	40	19724	0.79 [0.73–0.85]	0.61
OS	M2	40	19724	0.86 [0.77–0.96]	0.58
NRM	M1	12	1309	0.88 [0.66–1.18]	0.72
NRM	M2	12	1309	0.90 [0.59–1.36]	0.60

aGVHD II–IV	M1	28	4197	0.67 [0.59–0.78]	0.68
CMV	M1	22	3328	1.26 [1.07–1.47]	0.74
CMV	M2	22	3328	1.25 [1.01–1.54]	0.76
BSI	M1	6	806	1.87 [1.33–2.62]	0.96
IFI	M1	6	2243	0.43 [0.29–0.63]	1.64

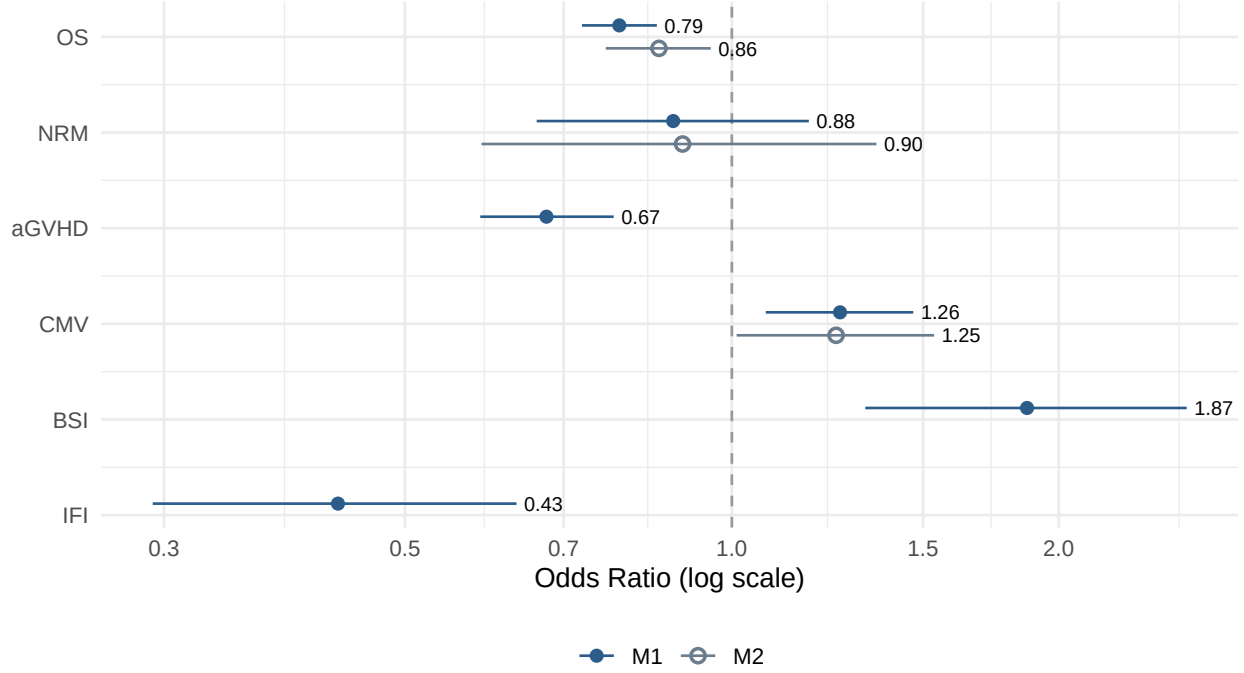


Figure 1: C1 Bayesian forest plot. Filled = M1 (direct); open = M2 (steroid-adjusted).

4.1 Pre vs post Block-9 comparison

Table 4: C1 M1 model comparison: pre-Block-9 vs post-Block-9.

Outcome	k (pre)	OR (pre)	k (post)	OR (post)	Shift
OS	35	0.71–0.83	40	0.79 [0.73–0.85]	+0.02
NRM	15	0.70–1.08	12	0.88 [0.66–1.18]	+0.02
aGVHD	24	0.54–0.74	28	0.67 [0.59–0.78]	+0.04
CMV	18	1.20–1.65	22	1.26 [1.07–1.47]	-0.15
BSI	6	1.85–3.10	6	1.87 [1.33–2.62]	-0.52
IFI	8	0.53–1.00	6	0.43 [0.29–0.63]	-0.30

C1 OS, NRM, and aGVHD are directionally stable (shifts = 0.04). C1 CMV attenuated from 1.41 to 1.26 but CrI still excludes null. BSI attenuated (2.39 → 1.87) and IFI strengthened substantially (0.73 → 0.43, though $I^2 = 1.64$ indicates extreme heterogeneity).

5 C2 results: PTCy vs ATG

Table 5: Comparison 2 primary results (post-Block-9).

Outcome	Model	k	N	OR [95% CrI]	
OS	M1	10	12851	0.81 [0.74–0.90]	0.51
OS	M2	10	12851	0.82 [0.71–0.94]	0.70
aGVHD II–IV	M1	9	1099	0.58 [0.44–0.77]	0.55
CMV	M1	13	1571	0.92 [0.73–1.15]	0.64
CMV	M2	13	1571	0.98 [0.78–1.24]	0.74

Note:

C2 NRM dropped below $k = 3$ after arm reclassification.

5.1 C2 sensitivity: expanded arm inclusion

The sensitivity reclassifies PTCy+ATG combination arms as “PTCy-side,” adding studies where the PTCy arm also used ATG.

Table 6: C2 primary vs sensitivity (expanded arm inclusion).

Outcome	Primary k	Primary OR	Sens k	Sens OR	Direction
OS	10	0.81 [0.74–0.90]	21	0.74 [0.66–0.83]	Strengthened
aGVHD	9	0.58 [0.44–0.77]	14	0.78 [0.61–0.98]	Attenuated
CMV	13	0.92 [0.73–1.15]	22	1.00 [0.85–1.18]	Converged to null

When PTCy arms also use ATG, both sides have deep T-cell depletion. CMV rates equalise ($OR \rightarrow 1.00$), aGVHD differential narrows (both arms depleting alloreactive T cells), and OS strengthens (redundant GVHD suppression benefits survival). This three-way pattern confirms the T-cell depletion hierarchy.

6 C3 results: within-PTCy dose/schedule variants

Table 7: Comparison 3 results (new). All CrIs cross null.

Outcome	k	N	OR [95% CrI]	
OS	6	4746	1.11 [0.95–1.30]	0.92
NRM	5	2529	0.96 [0.74–1.24]	1.22
aGVHD II–IV	7	2450	1.00 [0.80–1.25]	0.82
cGVHD mod/sev	4	383	0.67 [0.39–1.14]	0.41

All C3 models show null effects. The cGVHD estimate ($OR\ 0.67\ [0.39–1.14]$) shows a trend but $k = 4$ is insufficient. These comparisons (e.g., PTCy D+3/+4 vs D+3 only, reduced-dose vs standard-dose) require substantially more data.

7 CMV deep dive

7.1 Depletion gradient

The three-point gradient across primary and sensitivity analyses supports the depletion hierarchy more robustly than the pre-Block-9 binary reversal:

Table 8: CMV depletion gradient across analyses.

Analysis	Arms compared	k	OR	Interpretation
C1 primary	PTCy vs CNI	22	1.26 [1.07–1.47]	PTCy depletes more → more CMV
C2 primary	PTCy vs ATG	13	0.92 [0.73–1.15]	Near-parity
C2 sensitivity	PTCy+ATG vs ATG	22	1.00 [0.85–1.18]	Equal depletion → equal CMV

7.2 Serostatus subgroup analysis

Three studies reported CMV reactivation stratified by donor/recipient serostatus ($k = 1$ per D/R combination):

Table 9: CMV reactivation by D/R serostatus. All estimates are single-study (hypothesis-generating only).

D/R stratum	Source	PTCy %	Comp %	OR
D+/R+	Chorao 2024	47	40	1.34
D-/R+	Chorao 2024	67	69	0.91
D+/R-	Goldsmith 2021	20	11	2.06
D-/R-	Goldsmith 2021	2	1	1.89
R+ (pooled)	Goldsmith + Steiner	51–68	29–48	2.4–2.6

The D-/R+ stratum is the only one where PTCy did not increase CMV risk. In this configuration, the recipient has CMV-specific memory T cells (R+) but the donor does not (D-). PTCy’s selectivity for dividing cells may spare quiescent recipient-derived memory T cells while CNI-based regimens continuously suppress them.

7.3 Meta-regression

Neither recipient seropositivity ($k = 11$, $p = 0.72$) nor D+/R+ prevalence ($k = 8$, $p = 0.43$) significantly moderated the PTCy CMV effect as ecological study-level covariates. Both analyses were severely underpowered and restricted in range. Testing whether PTCy’s CMV effect truly differs by D/R serostatus requires **individual patient data** meta-analysis.

8 Publication bias

Table 10: Publication bias assessment for C1 outcomes with k = 10.

Outcome	k	Egger z	Egger p	T&F imputed	Original OR	Adjusted OR
OS	40	-1.02	0.314	7	0.77 [0.66–0.90]	0.86 [0.73–1.02]
aGVHD	28	-2.15	**0.041**	8	0.61 [0.47–0.79]	0.83 [0.61–1.13]
CMV	22	-0.21	0.835	0	1.25 [0.94–1.66]	1.25 [0.94–1.66]

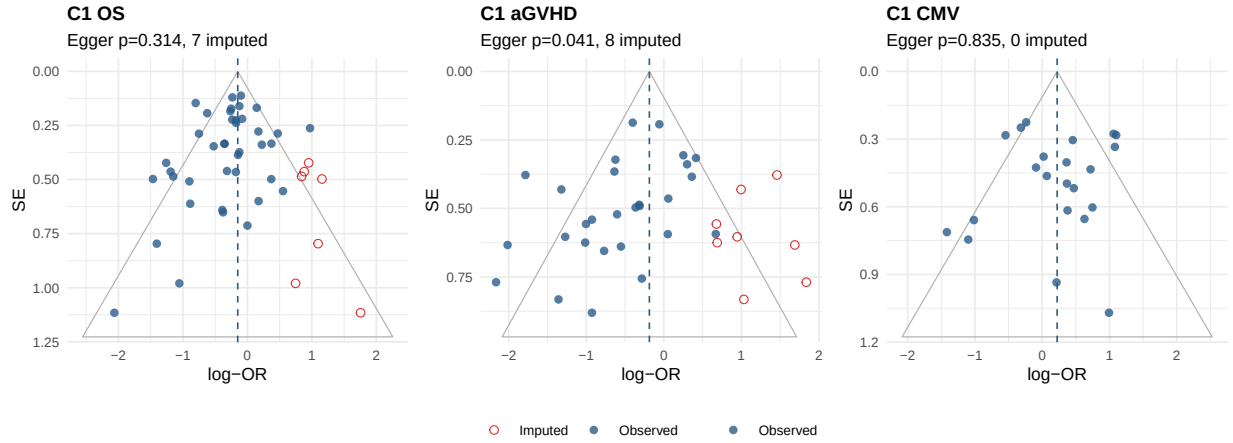


Figure 2: Funnel plots with trim-and-fill. Filled = observed; open red = imputed.

C1 aGVHD shows significant funnel asymmetry (Egger p = 0.041). Small studies disproportionately favour PTCy. The trim-and-fill adjusted OR (0.83 [0.61–1.13]) crosses null, suggesting the effect size may be partially inflated by selective reporting. The Bayesian estimate (OR 0.67) is less extreme than the frequentist 0.61 due to shrinkage but does not address selective non-publication.

C1 OS shows borderline concern: 7 studies imputed, adjusted OR 0.86 [0.73–1.02] touches null. Egger’s test is non-significant (p = 0.31). **C1 CMV** is symmetric with no imputed studies.

9 GRADE certainty summary

Table 11: GRADE certainty of evidence. Updated for post-Block-9 results.

Outcome	Comp	k	OR [95% CrI]	Certainty	Key limitation
OS	C1	40	0.79 [0.73–0.85]	LOW	I ² = 64%; obs only
OS	C2	10	0.81 [0.74–0.90]	LOW	I ² = 95%; attenuated from pre-B9
NRM	C1	12	0.88 [0.66–1.18]	VERY LOW	M2 mediation; CrI crosses null
NRM	C2	NA	— (k = 2)	—	Below minimum k
aGVHD II–IV	C1	28	0.67 [0.59–0.78]	VERY LOW	Egger p = 0.04; T&F adj 0.83
aGVHD II–IV	C2	9	0.58 [0.44–0.77]	LOW	I ² = 73%
CMV	C1	22	1.26 [1.07–1.47]	LOW	I ² = 62%

CMV	C2	13	0.92 [0.73–1.15]	VERY LOW	CrI crosses null
BSI	C1	6	1.87 [1.33–2.62]	VERY LOW	k = 6; freq CrI crosses null
IFI	C1	6	0.43 [0.29–0.63]	VERY LOW	= 1.64; extreme heterogeneity

Changes from pre-Block-9:

- C1 aGVHD: **LOW** → **VERY LOW** (publication bias detected)
- C2 NRM: **removed** (k = 2 after arm reclassification)
- C2 CMV: reasoning revised from “direction reversal” to “depletion gradient”
- C1 IFI: CrI now excludes null but = 1.64 warrants extreme caution

10 Remaining work

Task	Status
BMT director second-pass ROB spot-check	Not started
C3 website page and interpretation	Models fitted; no write-up
C1/C2 sensitivity (post-2020, haplo-only)	Pre-Block-9 models exist; not refit
RoBMA (Bayesian publication bias model)	Not attempted
Forest plots (study-level) for main outcomes	Not on website
cGVHD analysis for C1/C2	Not fitted
IPD meta-analysis for CMV × serostatus	Requires multi-centre collaboration